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By:

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Spinal cord segmentation in 3D MRI and predicting variations in cross-sectional area based on demographic factors

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CHAPTER 1:

| GENERAL INTRODUCTION

1. General Context

Medical imaging, particularly magnetic resonance imaging (MRI), has become an essential tool for the precise analysis of internal structures in neurodegenerative diseases. Neuromyelitis Optica (NMO) is an autoimmune disorder that primarily affects the optic nerve and causes atrophy of the spinal cord (SC). MRI plays a crucial role in detecting this SC atrophy by measuring the Spinal Cord Cross-Sectional Area (CSA), which is a common indicator of medullary atrophy in NMO patients.

2. Problematic

The manual analysis of MRI is time-consuming, subject to variability, and may lack consistency when handling extensive datasets. These challenges necessitate the development of automated, efficient, and accurate methods for the segmentation and analysis of SC images.

3. Overview on the Solution

This project proposes a solution based on a Convolutional Neural Network (CNN) architecture, specifically the U-Net, which is well-suited for image segmentation tasks. U-Net is designed to capture both global context and precise localization by combining a contracting path and an expansive path with skip connections.

4. Chapters' Presentation

Chapter 1: introduce the general and specific contexts, along with a detailed definition of the problem.
Chapter 2: review the state of the art in MRI segmentation, spinal cord morphology studies, and deep learning techniques.
Chapter 3: describe the materials and methods used for model development.
Chapter 4: present the results of segmentation and cross-sectional area analysis.
Chapter 5: discuss the findings, highlight the limitations, and propose future research directions.

CHAPTER 2:

| STATE OF THE ART

1. Relevant Work:

The automatic segmentation of the spinal cord (SC) and morphometric analysis of spinal MRI have been widely studied. De Leener et al. developed the Spinal Cord Toolbox (SCT 2017) [1], which includes the advanced segmentation algorithm sct_deepseg_sc based on U-Net convolutional neural networks for precise SC extraction from various MRI modalities. Cohen-Adad et al. (2014) [2] showed that the cross-sectional area (CSA) is a useful biomarker for monitoring SC atrophy in neurodegenerative diseases like multiple sclerosis. Bédard et al. (2023) [3] proposed a deep learning method for contrast-independent SC segmentation that reduces CSA variability by using "soft" segmentations accounting for partial volume effects. More recently, Boushila et al. (2024) [4] introduced a method to normalize CSA measurements using demographic factors (age, sex, height, weight) based on multiple linear regression to account for inter-subject variability in healthy populations.

2. Comparing Existing Work:

Table1: Comparison of Existing Studies

Criteria	De Leener et al(2017)	Toufani et al (2021) [5]	Bédard et al (2023)	Boushila et al (2024)	Classical Methods
Segmentation approach	U-Net	Semi-automated segmentation	U-Net	Not applicable	Thresholding, active contours
CSA calculating	included	Calculated at specific vertebral levels from masks	Calculated (via SCT)	SCT	None or manual analysis
Demographic variables used	Not integrated	Yes (age, sex, height, weight, BMI)	Yes (age, sex, height, weight	Yes (age, sex, height, weight)	Typically ignored
Accuracy & robustness of Segmentation	High accuracy	High accuracy	High accuracy	High accuracy	Low
Generalization	Moderate	Moderate	multi-center studies	multi-center studies	Poor

3. Criticism of Existing Work

Notwithstanding notable advancements, the existing systems have a number of drawbacks:

- ✓ **Absence Complete solutions:**

The majority of research divides analysis and segmentation. Segmentation, CSA extraction, and demographic-based prediction are not all handled by a single, streamlined pipeline.

- ✓ **Few research have used deep learning:**

To predict CSA based on demographic data, despite the fact that it is becoming more and more popular for segmentation. This could increase modeling flexibility and accuracy.

CHAPTER 3:

| CONTRIBUTION

In this project, we developed an automated pipeline that combines deep learning and regression analysis to study spinal cord morphology from MRI scans of healthy individuals. The goal of this pipeline is to allow fast, reproducible, and standardized analysis of the CSA, particularly at the C2–C3 level, while reducing manual intervention.

1. Dataset:

We used the Spine Generic dataset, a publicly available collection of 208 3D MRI data from healthy subjects. The scans were obtained from participants across 43 imaging centers, following a standardized acquisition protocol [6]. The MRI scans were performed using three different MRI scanner models, ensuring consistency across sites. The images are T2-weighted and acquired with an isotropic voxel size of

0.8 mm, providing high-resolution anatomical information suitable for quantitative analysis.

The SC was segmented using the SCT, and the dataset was split into training, validation, and testing sets following a standard 80/10/10 split. The masks generated by SCT served as ground truth for training the segmentation model.

2. Model training :

To segment the spinal cord automatically, we trained a convolutional neural network based on the U-Net architecture. U-Net is well-known for its effectiveness in medical image segmentation, thanks to its encoder-decoder structure and skip connections that help preserve spatial information

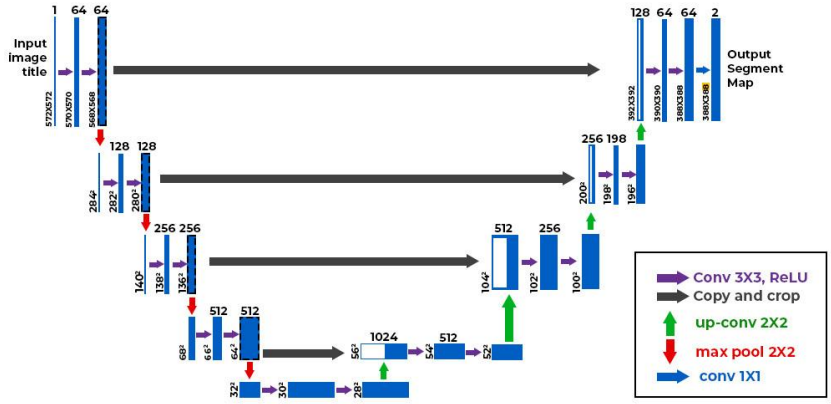


Figure 1-Architecture UNET

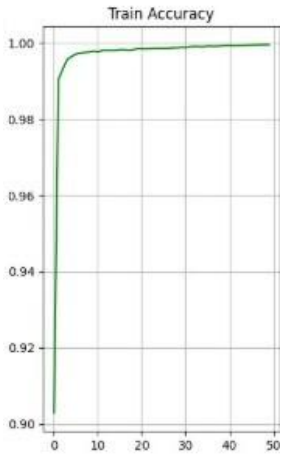


Figure 2-Train Accuracy

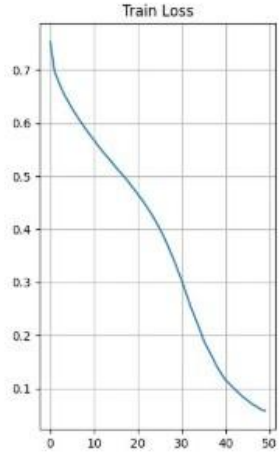


Figure2-Train Loss



Figure 4-Validation Accuracy

The model was trained in a supervised manner using binary masks as ground truth. We applied standard preprocessing techniques such as intensity normalization to improve the model's generalization. Training was carried out over 50 epochs with a batch size of 16, using the Adam optimizer and Dice loss as the objective function. The learning rate was fixed at 1e-4.

3. Cross-Sectional Area Computation:

Once the spinal cord masks were generated, we proceeded to compute the cross-sectional area at the C2–C3 vertebral level.

This region was identified using the sct_label_vertebrae command from SCT, which automatically labels vertebral levels. CSA was then calculated using sct_process_segmentation, a function that quantifies the area enclosed by the segmentation within a given vertebral segment. This provided us with reliable and anatomically meaningful measurements of spinal cord size.

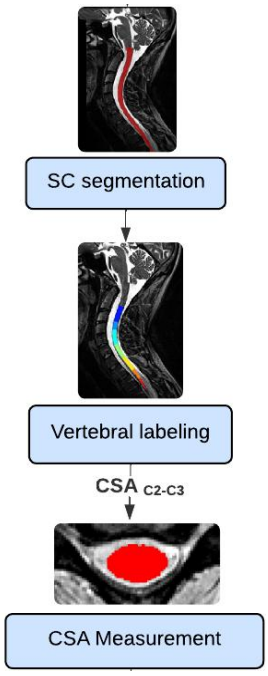


Figure 5- Spinal Cord Cross-Sectional Area(CSA)Measurement Pipelin

4. Modeling Demographic Influence on CSA:

To assess how demographic variables influence CSA, we used a multiple linear regression model inspired by the work of Boushila et al. (2024). This model helps quantify how factors such as age, sex, height, weight, and scanner type contribute to variations in CSA across individuals. By applying their normalization formula, we were able to adjust our CSA measurements to reduce inter-subject variability and better isolate the impact of individual demographic features. Since our pipeline was applied to the same dataset as theirs, we adopted their regression coefficients directly for consistency and comparability.

The regression formula used is as follows:

Formula:

$$CSA^{i}_{norm\ C2\text{-}C3} = CSA^{i}_{meas} + 4.3317\,(0.5316 - X^{i}_{,\ sex}) - 1.5694\,(0.5306 - X^{i}_{,\ manufacturer})$$

Where:

CSA^{i}_{meas} : measured CSA value from MRI.

$X^{i}_{,\ sex}$: encoded value for the subject's sex (0 = male, 1 = female).

$X^{i}_{,\ manufacturer}$: encoded value for the MRI scanner manufacturer

This normalization step allowed us to report CSA values that are corrected for sex and scanner-related variability, facilitating a more meaningful comparison across individuals.

Through this pipeline, we demonstrate that it is possible to automate the analysis of spinal cord morphology with minimal user interaction while maintaining scientific rigor and reproducibility.

CHAPTER 4:

| RESULTS AND DISCUSSIONS

5. Segmentation results:

The test results demonstrate strong performance of the segmentation model. With an average loss of 0.0747, the model shows effective learning and low error. The average Dice score of 0.8553 and IOU of 0.8742 indicate high overlap and precise segmentation of the spinal cord. Although the accuracy is very high (0.9992), it is less informative due to class imbalance. Overall, these metrics confirm the model's reliability for accurate spinal cord segmentation.

Table2: Performance Comparison of Deep Learning Models for Medical Image Segmentation

Model / Study	Application / Dataset	Dice Score	Accuracy	Loss
HOM Journal (2024)	Lung segmentation (CT scans)	0.90	High	Low
Focus U-Net (2021)	Polyp segmentation (colonoscopy)	0.878	High	Hybrid
ResUNet++ (2019)	Polyp segmentation (Kvasir-SEG)	0.813	High	Low
U-Net (ours)	MRI	0.8553	0.9992	0,074

Comparing to other methods, our model achieved satisfactory results, demonstrating an acceptable performance in delineating the anatomy of the spinal cord.



Figure 6-Image of segmentation results (MRI+mask)

1. CSA measurement results:

Table 3: résultats CSA moyenne des sujets

	Mean CSA(mm2)	Standard Deviation(mm2)
sub-nottwil01	65.7	1.79
sub-tokyoIngenia02	105.45	1.57
sub-tokoIngenia03	106.42	1.57

Normalization:

Table 4: CSA normalization results by subject – comparison between measured and predicted values

Subject	Measured CSA (mm^2)	Predicted CSA (mm^2)	Deviation (mm^2)	Normalized CSA
sub-nottwil01	105.54	50.2	55.34	0.476
sub-sherbrooke03	122.62	61.3	61.32	0.500
sub-tokyoIngenia02	105.45	49.0	56.45	0.465
sub-tokyoIngenia03	106.42	53.5	52.92	0.502

Application Interface:

The mobile application developed using Flutter includes three main interfaces: **Login, Registration**, and **Profile**.

The **Login** interface allows users to authenticate using their email and password, with a "Login" button to proceed. The **Registration** interface enables new users to sign up by providing a username, email, and password, and confirming via a "Sign Up" button.

Once registered, users access the **Profile interface**, where they can input demographic information such as age, sex, and height, and upload a MRI.

These data are sent to the backend server, which processes them to compute the CSA of the spinal cord and returns the result to the application

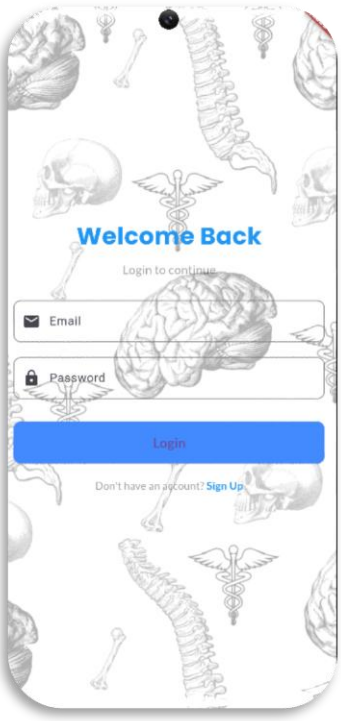


Figure 7-Main Interface

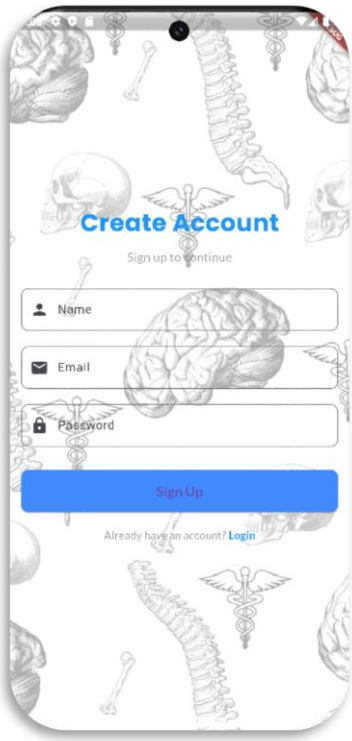


Figure 8-Register Interface



Figure9-Main Interface

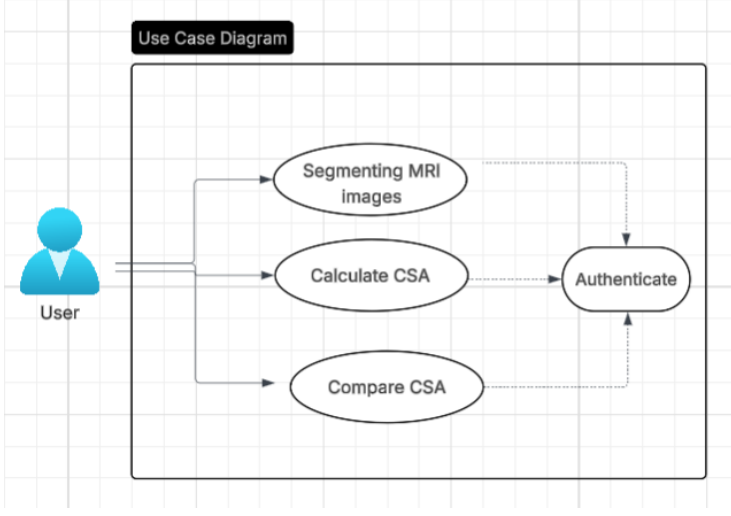


Figure 10-Use Case Diagramme

3. Analysis and Interpretation:

The applicability of the U-Net model for spinal cord segmentation from MRI was confirmed by its high segmentation accuracy (Dice0.2095).

The benefit of deep learning was highlighted by the failure of conventional techniques like thresholding in low-contrast areas.

A considerable amount of the variation in spinal cord shape can be explained by demographic factors, according to the CSA regression model.

The most significant factors were found to be height and sex. In keeping with established neurodegenerative tendencies, a small age-related reduction in CSA was noted.

The lack of significant multicollinearity or overfitting was suggested by the slight improvements in regularization that were seen when comparing linear and ridge regression.

These findings support the viability of automating and customizing spinal cord morphometry through statistical modeling and deep learning.

CHAPTER 5:

| CONCLUSIONS AND PERSPECTIVES

1. Summary of the Contribution:

In this work, we created a deep learning-based pipeline for automatically segmenting the spinal cord from T2-weighted MRI and then extracting the cross-sectional area (CSA).

The segmentation model was based on the U-Net architecture and performed well on data from healthy subjects.

We then used statistical regression models to examine the impact of demographic factors, specifically age, sex, height, and weight, on the morphology of the spinal cord using the extracted CSA values.

2. Highlight of the Results:

Our U-Net segmentation model achieved high accuracy, with a Dice 0.85, significantly outperforming traditional image processing methods.

CSA values extracted from the segmentation masks were found to vary meaningfully with demographic features. In particular, multiple linear regression showed that sex and height had the strongest impact on CSA values.

These findings are consistent with previous research and demonstrate the potential of automated pipelines for large-scale spinal cord morphometry studies.

3. Limitation of the Study:

Despite promising results, the study is limited by the homogeneity of the dataset, which consisted solely of healthy individuals and MRI scans from a single source. This restricts the generalizability of the findings to broader clinical populations. Additionally, the use of linear regression models may not fully capture complex relationships between demographic variables and CSA.

4. Future Work:

Future work could focus on extending the pipeline to pathological cases, such as patients with multiple sclerosis or spinal cord injury, to evaluate its diagnostic potential. Incorporating non-linear or machine learning-based regression models (e.g., random forests or neural networks) may improve CSA prediction accuracy. Additionally, expanding the dataset to include images from multiple scanners and institutions would allow better assessment of the generalizability and robustness of the segmentation model. Finally, integrating CSA normalization into clinical software tools could aid in personalized diagnostics and treatment monitoring.

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